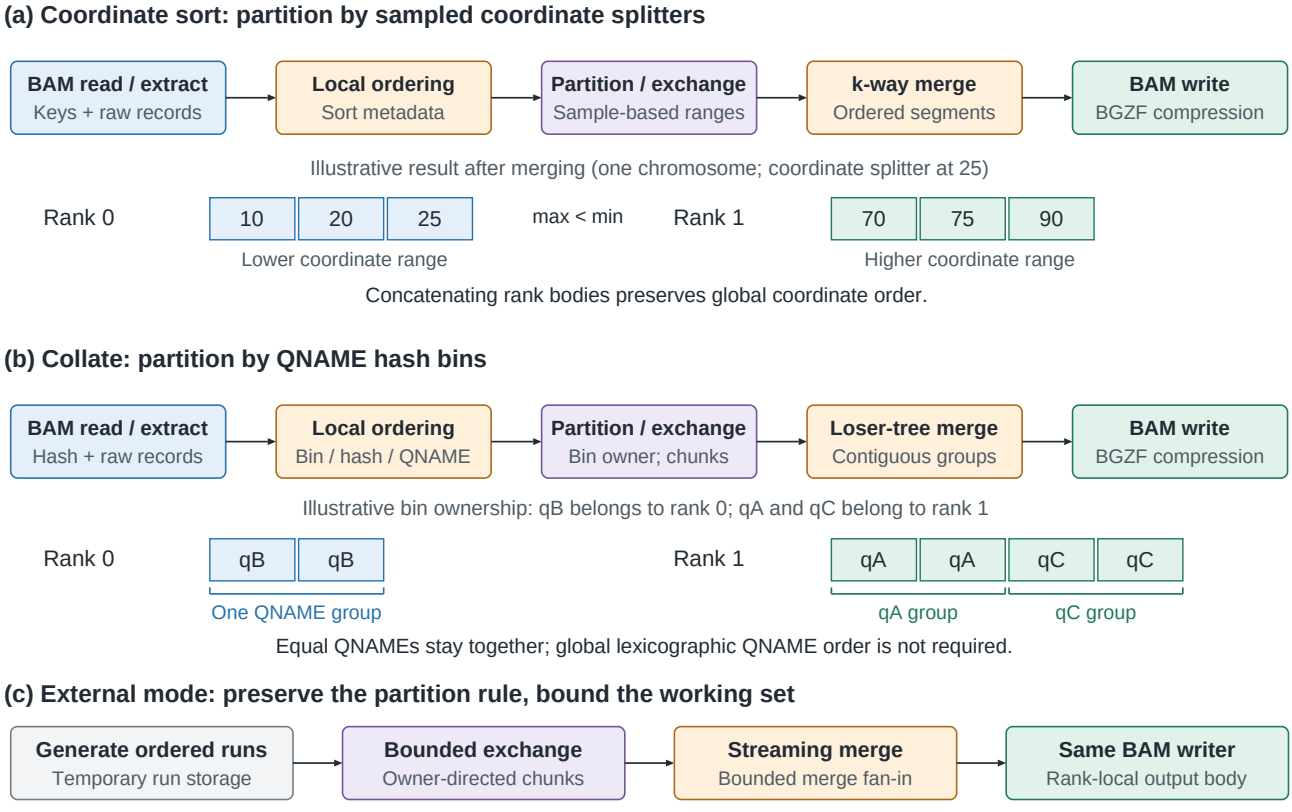
(a) Coordinate sort: partition by sampled coordinate splitters
BAM read / extract
Keys + raw records
Local ordering
Sort metadata
Partition / exchange
Sample-based ranges
k-way merge
Ordered segments
BAM write
BGZF compression
Illustrative result after merging (one chromosome; coordinate splitter at 25)
Rank 0
10
20
25
max < min
Rank 1
70
75
90
Lower coordinate range
Higher coordinate range
Concatenating rank bodies preserves global coordinate order.
(b) Collate: partition by QNAME hash bins
BAM read / extract
Hash + raw records
Local ordering
Bin / hash / QNAME
Partition / exchange
Bin owner; chunks
Loser-tree merge
Contiguous groups
BAM write
BGZF compression
Illustrative bin ownership: qB belongs to rank 0; qA and qC belong to rank 1
Rank 0
qB
qB
Rank 1
qA
qA
qC
qC
One QNAME group
qA group
qC group
Equal QNAMEs stay together; global lexicographic QNAME order is not required.
(c) External mode: preserve the partition rule, bound the working set
Generate ordered runs
Temporary run storage
Bounded exchange
Owner-directed chunks
Streaming merge
Bounded merge fan-in
Same BAM writer
Rank-local output body